

Statistical Analysis Plan (SAP)

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Title Relative Effectiveness of Social Media, Dating Apps, and Information Search Sites in Promoting HIV Self-testing: Observational Cohort Study

CRU/Department/Division/Center

IRB Number IRB #18-001580

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Original Creation Date

Version Date

Project Folder Location

Project Goal(s) Compare the relative effectiveness of social media sites, dating apps, and information search sites in promoting HIV self-testing among minority men who have sex with men (MSM) at an increased risk of HIV infection.

Submission Deadline(s) 1/28/2026

Effort Estimate (optional)

Investigator Agreement

- ☒ All statistical analyses included in an abstract or manuscript should reflect the work of the biostatistician(s) listed on this SAP. No changes or additional analyses should be made to the results or findings without discussing with the project biostatistician(s).
- ☒ All biostatisticians on this SAP should be given sufficient time to review the full presentation, abstract, manuscript, or grant and be included as co-authors on any abstract or manuscript resulting from the analyses.
- ☒ If substantial additional analysis is necessary or the aims of the project change, a new SAP will need to be developed.
- ☒ Publications resulting from this SAP are supported in part by the Duke CTSA and must cite grant number UL1TR002553 and be submitted to PubMed Central.
- ☒ I have reviewed the SAP and understand that any changes must be documented.

Acknowledged by: Trey Chase

Date: Click or tap to enter a date.

Activity Log

No wave of activity in wave 3 due to COVID 19.
Increased number of observations in data.

Acronyms

MSM	Men who have sex with men.
PrEP	Pre-Exposure Prophylaxis

1 Study Overview

Background/Introduction:

Advertisements on platforms such as social media sites, dating apps, and search engines are common ways for promotions to reach consumers. However, it remains unclear which platform is most efficient in promoting home HIV self-testing among minority men who have sex with men (MSM), given that users of various platforms may have different characteristics that impact their readiness for HIV testing. While these platforms have been widely adopted for HIV prevention recruitment, there is a lack of head-to-head comparisons evaluating their relative effectiveness in promoting HIV self-testing and facilitating test kit orders. Study Aims
[insert aims (this is in scientific terms)]

1.1 Study Hypotheses

1.1.1 Primary Hypotheses

There is an effect between various platforms and test kit order rate among MSM who are at high risk for HIV.

1.1.2 Secondary Hypotheses

- HIV self-test kit ordering differs by levels of substance use, including alcohol, cannabis, and other substances.
- HIV self-test kit ordering is associated with participants' stage of readiness for HIV testing.
- HIV self-test kit ordering is associated with attitudes toward HIV testing and HIV treatment.
- HIV self-test kit ordering is associated with HIV related stigma, including perceived stigma and social concerns related to HIV status.
- HIV self-test kit ordering is associated with levels of medical mistrust toward health care organizations and providers.
- HIV self-test kit ordering is associated with perceptions and beliefs related to HIV risk and consequences of HIV infection.
- HIV self-test kit ordering is associated with attitudes toward pre-exposure prophylaxis, including perceived barriers and facilitators to PrEP use.

2 Study Population

2.1 Inclusion Criteria

- Aged 18 to 30 years at enrollment
- Identified as a man who has sex with men
- Self-identified as Black or African American or Latinx, including multiracial or multiethnic individuals
- Reported condomless anal sex in the past 90 days or more than one male sexual partner in the past 90 days
- Resided in one of the study recruitment locations at the time of enrollment
- Able to provide electronic informed consent and complete online study procedures

2.2 Exclusion Criteria

- Self-reported HIV positive status at screening
- Reported HIV testing within 90 days prior to enrollment

- Current or recent use of pre-exposure prophylaxis, defined as PrEP use within the past 6 months
- Ineligible, duplicate, or invalid study entries identified during screening or data quality review

2.3 Data Acquisition

Fill in all relevant information:

Study design	Longitudinal observational cohort study evaluating online recruitment strategies for HIV self-testing
Data source/how the data were collected	Data were collected through web-based recruitment advertisements on social media, dating apps, and information search sites. Eligible participants completed an online eligibility screener, electronic consent, and baseline survey. HIV self-test kit ordering data were captured through automated reports from the test kit vendor and follow up surveys were administered electronically at 14 and 60 days.
Contact information for team member responsible for data collection/acquisition	Study principal investigator and research team at the University of California, Los Angeles (UCLA)
Date or version (if downloaded, provide date)	Data collected between January 2020 and September 2020Secure electronic transfer from online survey platform and automated vendor reports during the study period. Personally downloaded the data in January 2026.
Data transfer method and date	Secure electronic transfer from online survey platform and automated vendor reports during the study period.
Where dataset is stored	Access restricted institutional servers at UCLA.

3 Outcomes, Exposures, and Additional Variables of Interest

3.1 Primary Outcome(s)

Outcome	Description	Variables and Source	Specifications
Order HIV test kit	HIV self test kit ordering	ordered_test_kit (study tracking and	Binary variable coded as 1 = participant ordered an HIV self test kit within 60

		automated test kit vendor records)	days of code issuance, 0 = participant did not order a test kit within 60 days
Test Kit Order Rate	Rate of test kits ordered in the last 60 days for a specified platform.		Orders / 60

3.2 Secondary Outcome(s)

Outcome	Description	Variables and Source	Specifications
Substance Use	Use or desire of harmful substances within the last 3 months.		Q13_1 – Q13_24
Stage of HIV Testing	Time between survey and last HIV test.		Last HIV test interval
Attitudes towards HIV Testing and Treatment	General sentiment on HIV testing and treatment		Q15_1 – Q15_7
HIV-Related Stigma	Attitudes towards those with HIV.		Q14_2 – Q14_5
Medical Mistrust	Integrity/trustworthiness of medical institutions		Q16_1 – Q16_7

3.3 Additional Variables of Interest

Variable	Description	Variables and Source	Specifications
Do you self identify as	Race		Q5_3
Have you ever taken PrEP?	Medication, Behavioral		Q6_2
How often do you use condoms?	Behavioral, Sexual habits		Q11_3

4 Statistical Analysis Plan

Analyses were conducted to evaluate differences in HIV self-test kit ordering across recruitment platforms and to assess factors associated with kit ordering. Participants were summarized using descriptive statistics, with continuous variables reported using means and standard deviations or medians and interquartile ranges, and categorical variables reported as counts and percentages. The primary outcome, HIV self-test kit ordering, was summarized as daily order rates by platform and wave. Because ordering rates varied substantially by site and wave, comparisons were performed within waves rather than pooled across platforms, with pairwise comparisons adjusted for multiple testing. Secondary analyses compared demographic, behavioural, and attitudinal characteristics

between participants who ordered a test kit and those who did not, using appropriate parametric or nonparametric tests based on data type and distribution. Sensitivity analyses were conducted to assess the robustness of findings to alternative outcome definitions and study periods. All analyses were performed using R and the emmeans package, and statistical significance was evaluated at a two-sided α level of 0.05.

Table 1.

Summary of National Institute on Drug Abuse Clinical Trials Network Social Media Pre-exposure Prophylaxis Study, 2020, population sociodemographic and behavioral characteristics (N=254).

Characteristic	Value
Age in years, median (IQR)	25 (23-27)
Ethnicity, n (%)	
Hispanic/Latinx	66 (26)
Race, n (%)	
American Indian or Alaskan Native	1 (0.4)
Black or African American	196 (78.4)
White	28 (11.2)
Other	14 (5.6)
Multiracial	11 (4.4)
History of PrEP^a uptake, n (%)	
Never taken PrEP	232 (91.3)
In the past 6 months	22 (8.9)
Number of male sex partners in the past 90 days, median (IQR)	4 (3-6)
Condom use, n (%)	
Never	36 (14.2)
Sometimes	108 (42.5)
About half the time	37 (14.5)
Most of the time	68 (26.8)
Always	5 (2)
Condomless receptive anal sex in the past 90 days, n (%)	210 (82.7)
Ever tested for HIV during lifetime, n (%)	191 (75.2)
If tested for HIV, median (IQR)	
Months since last HIV test	11 (6-21)
If not tested for HIV, n (%)	63 (24.8%)
Main reasons cited by the 63 participants for not getting tested, n (%)	
Unlikely to be exposed to HIV	8 (12.7)
Afraid of testing HIV-positive	26 (41.3)
Did not want to think about HIV/HIV-positive	8 (12.7)
Worried about names being reported if positive	3 (4.8)
Dislike for needles	5 (8)
Unable to trust that the results will be confidential	3 (4.8)
Unaware of where to get tested	7 (11.1)
Other reasons	3 (4.8)

[Open in a new tab](#)

^aPrEP: pre-exposure prophylaxis.

4.1 Demographic and Clinical Characteristics ("Table 1")

4.2 Analyses Plan for Aim 1

[insert]

4.3 Analyses Plan for Aim 2

[insert]

ETC

5 Limitations

9 areas across observed individuals. Low enrollment amongst participants. Only chose most popular platforms. COVID 19 prevented the execution of wave 3.

6 Addendum for Additional Analyses

Software: R

7 Appendix

8 References

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Statistical Analysis Plan Checklist

Below you will find a checklist of recommended items to include in a statistical analysis plan. Some of these are specific to clinical trials (based on this [JAMA paper](#)) and some are other are specific to observational studies (based on [STROBE](#)/[RECORD](#) guidelines), so every item will not be necessary for every project. The biostatistician should start with the SAP template above and add in necessary information from the checklist. Item numbers that are starred (*) are not explicitly included in the SAP template and should be added by the author if relevant to the project. This checklist was developed using the [CONSORT 2010 Checklist](#).

Section/Topic	Item #	Description	Included (Yes/No/NA)
Administrative Information			
Study Information	1a	Descriptive title that matches the protocol, with SAP either as a forerunner or subtitle	_____
	1b	Trial registration number, protocol version number, and/or IRB number.	_____
	1c	CRU/Department/Division/Center/other collaborative unit that the study falls under	_____
Roles and responsibility	2a	Listing of principal investigators, clinical leads, and co-authors (if known)	_____
	2b	Name and affiliation of SAP author(s)	_____
	2c	Names, affiliations, and roles of other SAP contributors (e.g. senior statistician)	_____
SAP Information	3	SAP version number, with date of current version and original creation date	_____
Project Information	4a	Project folder location	_____
	4b	Project goals (e.g. manuscript, abstract, presentation, etc.)	_____
	4c	Project deadlines (of listed goals)	_____
	4d	Effort estimate	_____
Investigator Agreement			
Investigator Agreement	5	Confirmation that BERD Method Core's collaborative process has been reviewed, that all statistical analyses included in an abstract or manuscript should reflect the SAP, no changes should be made to the SAP without discussing with the SAP author, all biostatisticians on the SAP are co-authors on the manuscript, and that publications resulting from the SAP must cite grant number UL1TR002553 and be submitted to PubMed Central	_____
Signatures	6	Signatures of SAP author, senior statistician, and principal investigator(s)	_____
Activity Log			
SAP revisions	7a	SAP revision history with dates	_____
	7b	Justification for each SAP revision	_____
	7c*	Timing of SAP revision in relation to any interim analyses or submissions	_____

Study Overview

Background and introduction	8	Synopsis of scientific background and rationale for the study	
Aims and Hypotheses	9a	List of all scientific aims/objectives of the study, with specifications of primary, secondary, etc.	
	9b	List of all statistical hypotheses (corresponding to the scientific aims), with specifications of primary, secondary, etc.	
Variables of Interest	10a	List of all outcome/endpoint variables, with a description of their coding/units, timing, and source, corresponding to the statistical hypotheses. If any variables are defined using ICD or CPT codes, list them out.	
	10b	List of all exposure variables, with a description of their coding/units, timing, and source, corresponding to the statistical hypotheses. If any variables are defined using ICD or CPT codes, list them out.	
	10c	List of any additional variables of interest (e.g. covariates, potential confounders, effect modifiers, etc.) in the analysis	
	10d*	Location of data dictionary (or provided as an appendix)	
	10e*	Report category boundaries if continuous variables are collapsed into categories, and describe any other relevant data transformations	
Causal Graph	11*	May be helpful to include a DAG or other graph/diagram that describes the way the variables of interest are presumed to relate to each other	

Study Methods

Study Plan and Design	12a	Description of the study design (e.g. parallel group randomized trial, case-control study, cohort study, etc.)	
	12b*	Study setting, location, and relevant dates (e.g. periods of enrolment, exposure, follow-up, and collection)	
	12c*	Description of intervention or exposure groups, with allocation ratios, and details of any matching criteria	
	12d*	Details on randomization (e.g. stratification factors) and blinding procedures	
	12e	List of eligibility and/or inclusion/exclusion criteria	
	12f*	Description of screening/enrolment/recruitment processes	
	12g*	Description of patient flow (e.g. CONSORT diagram)	
	12h*	Description of analysis population (e.g. intention to treat, per protocol, etc.)	
	12i*	Definitions of adherence/compliance, protocol deviations, loss-to-follow-up, adverse events, etc.	
	12j*	Time points at which outcomes are measured	

	12k*	Timing of final analyses (are all outcomes analysed collectively, or will short-term outcomes be analysed separately from long-term outcomes, etc.)	
Sample Size	13a*	Sample size calculation or justification (either provided in full or summarized, with link to original source)	
	13b*	Description of pre-planned subgroup analyses, power for these analyses, and planned multiple comparison adjustment procedures	
Interim Analyses	14a*	Description of what interim analyses will be conducted at which time points, and what methods used to adjust significance levels due to the interim analysis	
	14b*	Details of any guidelines (e.g. safety, futility) for stopping the study early	
	14c*	Details of any changes to trial design due to interim analyses (e.g. enrolling more patients)	
Data	15a	Description of data collection/acquisition process, with contact information for team member responsible	
	15b	Description of data flow/transfer from primary data collection through to creation of final analysis dataset	
	15c	Data transfer method and date	
	15d	Folder location where datasets are stored	
	15e*	Description of any additional data management, quality control, or processing undertaken	
	15f*	If any data are extracted from a database, a description of the database and the query used for the extraction, and whether/how it was merged with any data from outside that database. If the study involved linkage of databases, consider use of a flow diagram to demonstrate the data linkage process, including the number of individuals with linked data at each stage.	
	15f*	Description of any other data sources incorporated in the analysis	
Missing Data	16a*	Description of sources and magnitudes of missing data	
	16b*	Description of how missing data patterns will be presented/summarized (may be helpful to have a table shell or draft CONSORT-style diagram)	
	16c*	Description of contingency plans for handling missing data in analysis	
Simulations	17a*	If conducting a simulation, a description of the purpose of the simulation and its design (e.g. fully factorial, partially factorial, grid search, etc.)	
	17b*	Define the fixed and variable factors or parameters in the simulation, the estimands/targets of the simulation, and the performance measures to be estimated (with justifications of their relevance to the estimands/targets)	
	17c*	Description of the tabular and graphical presentations of simulation results and their interpretation	

Statistical Analysis Plan

Statistical Significance	18a*	Hypothesis testing framework (e.g. superiority, equivalence, non-inferiority), or description of alternative analytic framework (e.g. evaluation of a posterior in a Bayesian analysis, etc.)	
	18b*	Level of significance for primary hypotheses, including a description and rationale for any multiple comparisons adjustment or Type I error control procedures	
	18c*	Description of any decision-making rules based on confidence intervals, credible intervals, prediction intervals, Bayes' factors, or other alternative inferential methods	
	18d*	Description of how the results of any hypothesis tests (or alternative inferential methods) will be interpreted with respect to both the statistical hypotheses and scientific aims/objectives of the study	
Descriptive Statistics	19a*	List of characteristics (e.g. demographic, clinical) to be summarized descriptively (e.g. "Table 1")	
	19b*	Description of how these characteristics will be summarized descriptively (e.g. means/medians vs. N (%), tabular displays, graphical displays, etc.)	
	19c*	Summarize follow-up time (e.g. average and total amount) and number of events	
Analysis Methods	20a	For each aim/hypothesis (see items 9a/9b), a description of what analysis method will be used and how the results from this method will be reported and interpreted	
	20b*	Description of any transformations, standardizations, covariate or confounder adjustments, weighting, or stratification methods to be used and why.	
	20c*	For each analytic method proposed, a description of the assumptions of that method and what processes will be used to evaluate whether or not those assumptions hold	
	20d*	Details of contingency plans/alternative methods to be used if the assumptions are found not to hold	
	20e*	In the case of non-standard test statistics, formulas provided for the test statistic with a description of the mathematical null hypothesis, how significance is determined, and how the test statistic is interpreted	
	20f*	In the case of regression models, formulas provided for the full model with a description of which parameters are to be used, how they will be interpreted, how confidence intervals will be constructed, etc.	
	20g*	In the case of survey, hierarchical/nested, or clustered data, a description of what methods will be used to adjust for the data structure and why (e.g. if using a GEE, describing which correlation structure and why it was chosen, etc.)	
	20h*	For non-continuous outcomes, clearly explain the effect used (e.g. risk difference, risk ratio, odds ratio, etc.), whether it is relative or absolute, and justify why that was chosen as the effect measure of interest	
	20i*	Documentation of any non-standard methods used (e.g. using alternative degree of freedom calculation methods, using a non-canonical link function, etc.)	

	20j*	Description of any limitations, sources of bias, internal/external validity, and other relevant discussions concerning the interpretation and generalizability of the design or methods used	
Additional Analysis Methods	21a*	Description of any pre-planned sensitivity analyses and how they will be interpreted	
	21b*	Description of pre-planned subgroup analyses, power for these analyses, and planned multiple comparison adjustment procedures	
	21c*	Description of any additional post-hoc calculations or analyses (e.g. evaluating interaction/modification effects, calculating mediation or local average treatment effects, evaluation of AUROC curves, etc.)	
	21d*	If conducting any bootstrap analyses, a description of the sampling algorithm and number of iterations used	
	21e*	If conducting any cross-validation procedures, a description of how the cross-validation is conducted (e.g. leave-one-out, train/validation/test, etc.)	
Exploratory Analyses	22a*	Description and justification for any pre-planned exploratory analyses and what methods will be used to conduct them	
	22b*	Framework for conducting any unplanned exploratory analyses and how they will be integrated into the planned analysis	
Software	23*	List of statistical software (along with version numbers) to be used for each phase of the analysis; in the case of R or Stata, additionally list any requisite installed packages and their version numbers	
Other	24*	Description of any additional planned analyses of the data (e.g. a safety analysis looking at adverse event rates for a Data Safety Monitoring Board, etc.)	
Tables and Figures			
Table Shells	25*	Example tables related to any of the conducted analyses; if possible including any available preliminary data	
Example Figures	26*	Example figures related to any of the conducted analyses; if possible including any available preliminary data.	
References			
References	27a	References for any non-standard statistical methods used	
	27b	References (and locations) for any relevant protocols, standard operating procedures, or other documents cited in the SAP	
Additional Information			
Appendices	28*	If necessary, appendices may be included (e.g. a full data dictionary, a copy of a Case Report Form, etc.)	
Addendums	29*	Any additional analyses conducted that were not included in the SAP should be documented in an addendum, describing the purpose of the additional analysis, when it was conducted, and by whom	